

Bachelor's Degree in Mathematics and Computing, 2025–26 Information Theory. First Midterm.

INSTRUCTIONS:

- Write your surname followed by your name in uppercase letters on the line below.
- All papers you use, whether written on or not, must include your name.
- Exams are official documents. Exams written in pencil will not be marked.
- You must submit your exam **immediately** if you allow others to see it.
- The use of books, notes, calculators, or any other aids is not permitted.
- Mobile phones must not be used or kept within reach, even if turned off.

Surname Name

Questions (0.5 points each, 2.5 points total)

Are the following statements true or false? Briefly justify your answers, citing relevant facts/theorems if true, or giving a counterexample if false:

- Q1.-** If $H(X) = 0$ for a discrete random variable X , then X is almost surely constant.
- Q2.-** If $H(X | Y) = 0$, then there exists a function f such that $X = f(Y)$ holds with probability 1.
- Q3.-** If $I(X; Y) = 0$, then $H(X | Y) = H(X)$.
- Q4.-** For all probability distributions P and Q on the same alphabet, the KL divergence is symmetric: $D(P \| Q) = D(Q \| P)$.
- Q5.-** If integers $\ell_1, \dots, \ell_m$ satisfy $\sum_{i=1}^m 2^{-\ell_i} \leq 1$, then there exists a *prefix-free* binary code whose codeword lengths are exactly $\ell_1, \dots, \ell_m$.

Problems

P1.- [2 points] Let (X, Y) be binary random variables with joint distribution

$P_{X,Y}(x, y)$	$y = 0$	$y = 1$
$x = 0$	$\frac{1}{2}$	$\frac{1}{4}$
$x = 1$	0	$\frac{1}{4}$

- Compute $H(X)$, $H(Y)$, $H(X, Y)$, and $H(X | Y)$.
- Compute $I(X; Y)$.

P3.- [2 points] Consider a source X with alphabet $\mathcal{X} = \{a, b, c, d\}$ and probabilities

$$P_X(a) = \frac{1}{2}, \quad P_X(b) = \frac{1}{4}, \quad P_X(c) = \frac{1}{8}, \quad P_X(d) = \frac{1}{8}.$$

- Compute the entropy $H(X)$.
- Consider the proposed binary code lengths

$$\ell(a) = 1, \quad \ell(b) = 2, \quad \ell(c) = 3, \quad \ell(d) = 3.$$

Use Kraft's inequality to decide whether there exists a prefix-free code with these lengths.

- If it exists, explicitly construct one such prefix-free code and compute its average length $\ell_C(P_X)$.
- Verify that your average length is consistent with Shannon's source-coding theorem.

P4.- [2 points] Let X take values in $\{a, b, c, d, e\}$ with probabilities

$$P_X(a) = 0.30, \quad P_X(b) = 0.25, \quad P_X(c) = 0.20, \quad P_X(d) = 0.15, \quad P_X(e) = 0.10.$$

- Construct a *binary Huffman code* for this source (give codewords).
- Compute the average codeword length and compare it to $H(X)$ and $H(X) + 1$.

P6.- [1.5 point] Let $C : \mathcal{X} \rightarrow \{0, 1\}^*$ be a binary prefix-free code.

Prove that C is uniquely decodable, i.e., show that the extension

$$C^* : \mathcal{X}^* \rightarrow \{0, 1\}^*$$

defined by concatenation is injective.